their body temperature at a more or less constant level, insects have a variable body temperature that changes with their environment.

In the early 19th century, however, British entomologist George Newport discovered that some moths and bees raise the temperature of their thorax (the central part of the body, to which wings and limbs attach) above that of the surrounding air by rapidly flexing their muscles. It is now known that many insects are heterotherms, maintaining different temperatures in different parts of the body, and are sometimes far warmer than the ambient temperature.

“In insects… the active flight muscles… are, metabolically, the most active tissues known.”

Bernd Heinrich

The right temperature

The main challenge facing insects is how to get warm enough to fly but cool enough not to overheat. German–American entomologist Bernd Heinrich explained in 1974 how moths, bees, and beetles could continue to function by controlling their own temperature. He realized that insects’ thermal adaptations do not differ as much from those of vertebrates as had been thought.

Most flying insects have higher metabolic rates than other animals but their small body size means they lose heat rapidly, so they cannot keep their temperature constant at all times. The minimum temperature that allows an insect to fly varies from species to species, but the maximum temperature falls within 104–113°F (40–45°C). To prevent overheating, insects can transfer heat from the thorax to the abdomen.

Many larger flying insects would remain grounded if they were not able to increase the temperature of their flight muscles. These insects “quiver” the muscles that control the upbeat and downbeat of the wings to generate